

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 13

BIOTECHNOLOGY - 3

BIOTECHNOLOGY

DNA and RNA

- The nucleus contains chromosomes which are made up of DNA and proteins.
- Heterocyclic compounds – It is a cyclical compound with a ring like structure.
- Heterocyclic compounds are – Adenine, Guanine, Uracil, Cytosine and Thymine. These are also known as nitrogenous bases.
- When heterocyclic compounds are attached to sugar, they are called Nucleosides.
- If a phosphate compound is also attached, then it is called Nucleotide.
- So, a Nucleotide consists of – Heterocyclic compound, one Sugar and one Phosphate.
- Nucleic Acids consist of Nucleotides. Nucleic acids are polymers of Nucleotides.
- Nucleic acids are responsible for the transfer of characteristics from parents to offspring.

BIOTECHNOLOGY

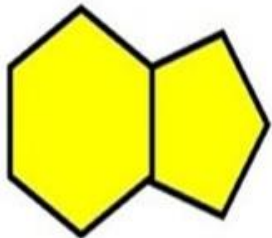
DNA and RNA

- There are two types of Nucleic acids – DNA and RNA.
- Nucleic acid containing Deoxyribose is called Deoxyribose Nucleic Acid (DNA) and that which contains Ribose is called Ribonucleic Acid (RNA).
- Both DNA and RNA contain Adenine, Guanine and Cytosine. The fourth compound is Thymine in DNA and Uracil in RNA.

Found in:

DNA

RNA

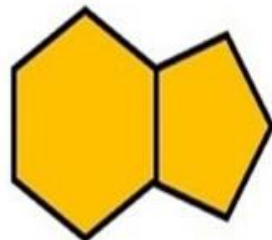


Guanine

Found in:

DNA

RNA

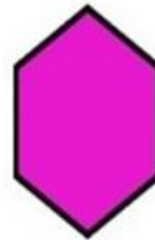


Adenine

Found in:

DNA

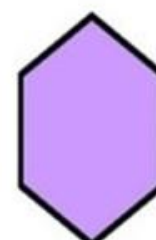
RNA



Cytosine

Found in:

DNA



Thymine

Found in:

RNA

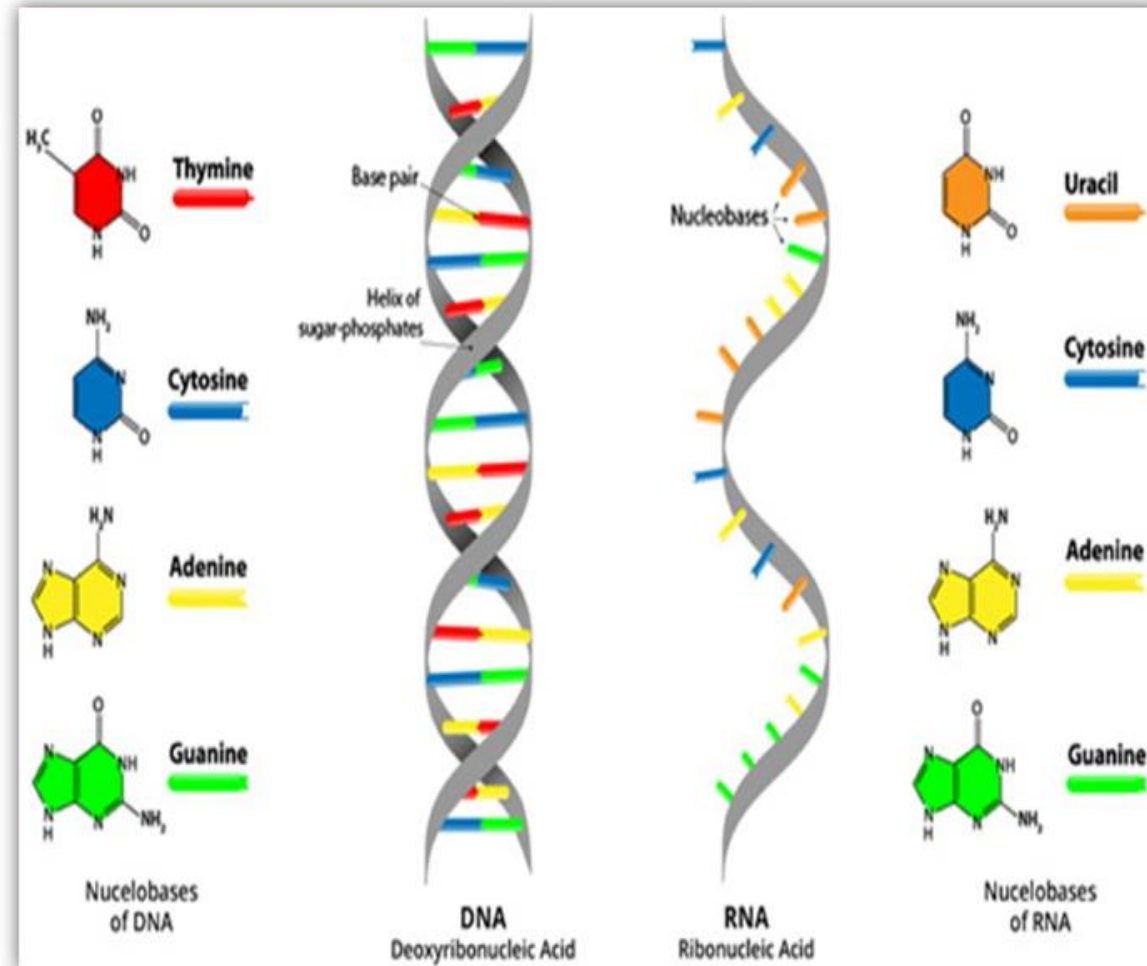


Uracil

BIOTECHNOLOGY

DNA and RNA

- The structure of DNA is double helix, while that of RNA is single.
- DNA is the reserve of genetic information.
- DNA contains coded information about the proteins to be synthesized inside the cell.
- RNA carries out the protein synthesis in the cell based on the coded information about proteins provided by DNA.
- So, RNA has the function of decoding and expression of genetic



GENETIC ENGINEERING

- It involves the techniques to alter the chemical composition of genetic material (DNA and RNA) and thus change the phenotype of the organism.
- Phenotype – A set of observable characteristics of an individual.
- Asexual reproduction preserves the genetic information while sexual reproduction permits variation.
- Traditional hybridisation procedures used in plant and animal husbandry led to inclusion and multiplication of undesirable genes along with the desirable genes.
- The techniques of Genetic Engineering like Recombinant DNA Technology, Gene Cloning etc, overcome this limitation and allow us to isolate and introduce only one or a group of desirable genes without introducing undesirable genes into the target organism.
- There are three basic steps in genetically modifying an organism –
 1. Identification of DNA with desirable genes.
 2. Introduction of the identified gene into the host.
 3. Maintenance of the introduced DNA in the host and transfer of the DNA to its progeny.

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